Zihan Yang

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Research Profile

Researcher at the intersection of control and machine learning, focusing on robust locomotion, disturbance estimation, and feedback-informed robot learning. Currently a research intern at Galbot working on humanoid locomotion and manipulation. Work includes two ICRA 2026 papers, an accepted *npj Robotics* article, an ICLR 2025 oral paper, and the ICCA 2024 Best Student Paper Award.

Education

Beihang University Sep 2023–Jan 2026
M.S. in Aerospace Science and Technology

- GPA: 3.80/4.00; supervised by Prof. Kexin Guo, Prof. Xiang Yu, and Prof. Lei Guo.

Northwestern Polytechnical University / Queen Mary University of London Sep 2019–Jun 2023
Joint NPU–QMUL B.Eng. dual-degree program in Materials Science and Engineering

- NPU GPA: 3.59/4.00; QMUL degree classification: First Class Honours.

Research Experience

Galbot Present
Research Intern

- Conduct research on humanoid locomotion and manipulation.

Beihang University Sep 2023–Jan 2026
Graduate Researcher, control and machine learning

- Developed learning-based disturbance estimators: a feedback-calibrated meta-learning framework demonstrated in quadrotor flight and a Chebyshev-based observer that decomposes coupled internal and external disturbances.
- Co-developed feedback neural networks that improve the generalization of neural ODEs; validated them on real-object trajectory prediction and quadrotor model predictive control.
- Developed TRACE, a closed-loop trajectory-refinement method for safe and accurate quadrotor maneuvers; received the ICCA 2024 Best Student Paper Award.

Beihang University / Unitree Robotics 2025
Quadruped robotics research

- Developed an observer-enhanced locomotion policy and worked on LiDAR-inertial odometry and mapping for quadruped robots.

Beihang University / China Academy of Launch Vehicle Technology 2024
Quadrotor swarm planning and control

- Optimized collision-free swarm trajectories with successive convexification and designed disturbance-observer-based tracking control for operation under wind disturbances.

Northwestern Polytechnical University 2021–2023
Undergraduate research

- Coupled porosity, heat-conduction, and stress models to study sintering behavior in large porous 3D-printed alumina components for the undergraduate thesis.
- Trained a ResNet-based inverse-design model on 10,000 airfoils with XFOIL-computed aerodynamic properties.

Publications

* Equal contribution.

Peer-reviewed publications

[1] J. Jia*, S. Han*, M. Wang*, G. Li, **Z. Yang**, S. Zhou, K. Guo, J. Yang, X. Yu, W. Wang, and L. Guo, “Physics filtering favors the generalization of robot learning,” *npj Robotics*, 2026, Accepted.

[2] **Z. Yang**, J. Jia, M. Wang, Y. Liu, K. Guo, and X. Yu, “Unified meta-representation and feedback calibration for general disturbance estimation,” in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[3] J. Jia*, M. Wang*, **Z. Yang**, B. Yang, Y. Liu, K. Guo, and X. Yu, “Learning-based observer for coupled disturbance,” in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[4] K. Guo, Y. Liu, J. Jia, **Z. Yang**, S. Zhou, X. Yu, and L. Guo, “Flying in harsh environments: Anti-rain quantification system and control strategy,” *IEEE Transactions on Industrial Electronics*, vol. 72, no. 10, pp. 10 349–10 358, 2025.

[5] X. Song, X. Guo, G. Liu, **Z. Yang**, and L. Liu, “Co-optimization of motion and energy domain for hydrogen-powered hybrid UAVs: A bi-directional coupling architecture,” in *2025 IEEE 19th International Conference on Control & Automation (ICCA)*, 2025, pp. 878–884.

[6] J. Jia*, **Z. Yang***, M. Wang, K. Guo, J. Yang, X. Yu, and L. Guo, “Feedback favors the generalization of neural ODEs,” in *The Thirteenth International Conference on Learning Representations (ICLR)*, Oral paper (1.8% oral acceptance rate), 2025.

[7] **Z. Yang**, J. Jia, Y. Liu, K. Guo, X. Yu, and L. Guo, “TRACE: Trajectory refinement with control error enables safe and accurate maneuvers,” in *2024 IEEE 18th International Conference on Control & Automation (ICCA)*, Best Student Paper Award, 2024, pp. 154–161.

Manuscripts and preprints

[8] **Z. Yang**, S. Han, K. Guo, and X. Yu, “CARO: Contact-agnostic residual observation for zero-shot robust quadruped locomotion,” 2026, Manuscript in preparation.

[9] K. Guo*, **Z. Yang***, Y. Liu, J. Jia, and X. Yu, “Optimizing control-friendly trajectories with self-supervised residual learning,” 2026, Under review.

Honors and Awards

Oral Paper (1.8% Oral Acceptance Rate) Apr 2025
The Thirteenth International Conference on Learning Representations (ICLR), Singapore

Best Student Paper Award Jun 2024
IEEE 18th International Conference on Control and Automation (ICCA), Reykjavík, Iceland

Second-Class Scholarship 2024, 2023
Beihang University

Technical Skills

Programming: Python; C/C++.

Robotics/ML: Learning-based control; neural ODEs; meta-learning; LiDAR-inertial odometry/mapping.

Simulation/Control: MATLAB/Simulink; XFOIL; trajectory optimization; disturbance observers; LQR; PID.

Hardware: Fixed-wing, quadrotor, and quadruped systems; Pixhawk, Jetson Nano, and Intel NUC; 3D printing, laser cutting, and electronic assembly.

Languages: Mandarin Chinese (native); English (fluent, IELTS 7.5); Japanese (basic).